KEEPER

Urban Investment & Resilience Intelligence

LAHORE FLOOD RESILIENCE INVESTMENT BRIEF

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Executive Summary

Given a budget of PKR 100,000,000, KEEPER identifies the combination of zone-level interventions that maximizes modeled public benefit. The recommended portfolio commits PKR 98,200,000 (1,800,000 remaining), reaching an estimated 379,435 people and reducing modeled flood risk by approximately 7.7% across the portfolio's zones.

Priority Areas

- Green Town (Z16) - risk class: HIGH
- Chah Miran (Z21) - risk class: HIGH
- Ravi Road (Z14) - risk class: HIGH
- Anarkali (Z22) - risk class: HIGH
- Harbanspura (Z08) - risk class: HIGH
- Mughalpura (Z12) - risk class: MODERATE
- Sabzazar (Z26) - risk class: HIGH
- Garhi Shahu (Z23) - risk class: HIGH
- Walton (Z17) - risk class: MODERATE
- Thokar Niaz Baig (Z28) - risk class: HIGH

Recommended Interventions

1. Green Town (Z16) - Flood Barrier - PKR 22,500,000 - benefit score 34.0
2. Chah Miran (Z21) - Flood Barrier - PKR 22,500,000 - benefit score 29.1
3. Ravi Road (Z14) - Drainage Upgrade - PKR 12,400,000 - benefit score 16.1
4. Anarkali (Z22) - Drainage Upgrade - PKR 12,400,000 - benefit score 15.4
5. Harbanspura (Z08) - Drainage Upgrade - PKR 12,400,000 - benefit score 15.0
6. Mughalpura (Z12) - Drain Maintenance - PKR 3,200,000 - benefit score 3.7
7. Sabzazar (Z26) - Drain Maintenance - PKR 3,200,000 - benefit score 3.5
8. Garhi Shahu (Z23) - Drain Maintenance - PKR 3,200,000 - benefit score 3.4
9. Walton (Z17) - Drain Maintenance - PKR 3,200,000 - benefit score 3.2
10. Thokar Niaz Baig (Z28) - Drain Maintenance - PKR 3,200,000 - benefit score 3.1

Budget Allocation

Total budget: PKR 100,000,000
Allocated: PKR 98,200,000
Remaining: PKR 1,800,000

Expected Benefits

Estimated population benefit: 379,435 residents
Estimated risk reduction: 7.7%

Methodology

Zone flood risk is a transparent weighted sum of flood exposure, drainage vulnerability, rainfall risk, and infrastructure vulnerability (0-100 scale). Population exposure combines risk with population count. For each zone-intervention pair, a benefit score is computed as population exposure x intervention effectiveness x risk reduction. Google OR-Tools solves a 0/1 knapsack integer program to select the combination of interventions that maximizes total benefit within the available budget, subject to one intervention per zone and per-intervention implementation-capacity limits.

Data Sources

This brief was generated using a demonstration dataset for Lahore. See data/README.md in the KEEPER repository for a full description of what is synthetic versus sourced.

Assumptions & Limitations

Intervention costs, risk-reduction percentages, and effectiveness figures are illustrative planning assumptions, not sourced engineering estimates. Zone-level risk and population figures in the current dataset are synthetic. These results are decision-support estimates, not engineering recommendations, and should be validated against sourced data and professional review before informing real capital allocation.

Recommended Next Steps

1. Replace synthetic zone data with sourced hazard, drainage, and population data.
2. Validate intervention cost and effectiveness assumptions with engineering/planning partners.
3. Re-run optimization against validated data and compare results.
4. Pilot the top-ranked intervention in the highest-priority zone.